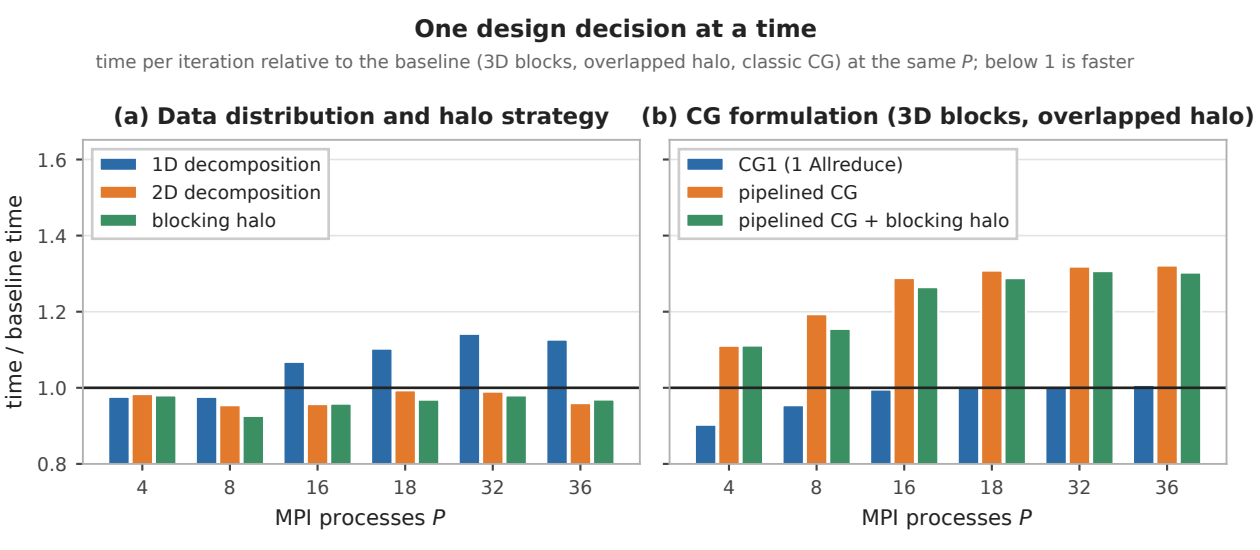
One design decision at a time
time per iteration relative to the baseline (3D blocks, overlapped halo, classic CG) at the same P; below 1 is faster
(a) Data distribution and halo strategy
(b) CG formulation (3D blocks, overlapped halo)
time / baseline time
1.6
1.4
1.2
1.0
0.8
1D decomposition
2D decomposition
blocking halo
CG1 (1 Allreduce)
pipelined CG
pipelined CG + blocking halo
4
8
16
18
32
36
MPI processes P
MPI processes P